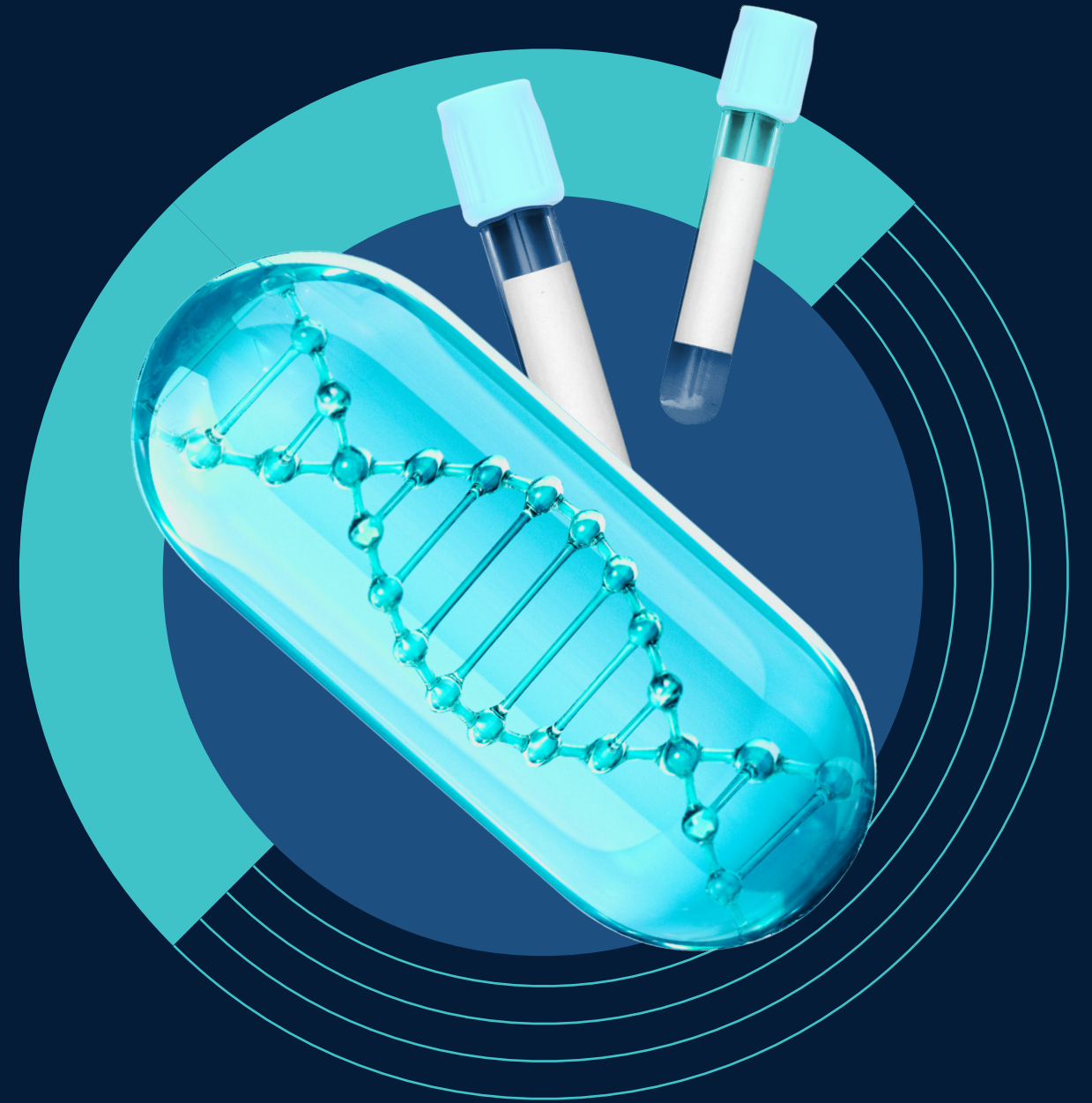


 EMERGING TECH RESEARCH

Pharma Biotech VC Trends

VC activity across the pharma biotech ecosystem

Q1
2025





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For previous updates as well as our complete biotools research, please see the designated [analyst workspace](#) on the PitchBook Platform.



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Pharma biotools landscape

- 1** Mature modalities
- 2** Advanced modalities
- 3** Emerging modalities





Pharma biotools VC ecosystem market map

This market map is an overview of venture-backed or growth-stage companies that have received venture capital or other notable private investments as of Q4 2024.

[Click to view the latest interactive map on the PitchBook Platform.](#)

1 Mature

Analytical & pharma tools

biodesix

908devices

AMPHASTAR

WuXiDiagnostics

Snibe

UNLEARN

ZEPHYR AI

Oncimmune

ibp

AZI

Basic molecular & cell biology

Adaptive

natera

GenScript

Vazyme

ACFO

OWKIN

scipher

DIAGENS

OpGen

NURITAS

Bioinformatics & cheminformatics

BIOSSENS

lucidata

IMiDomics

KNOWN MEDICINE

NeoTerreX

Traditional manufacturing, bioprocessing & pharma services

Formation Bio

FirGenix

QIVA

THOUSAND OAKS

Asymchem Bio

chime

神州细胞

JOINN

ESCO

ascend

2 Advanced

Genomics & sequencing

Oxford Nanopore Technologies

MGI

HUMAN LONGEVITY

DNAexus

genuity

DELFI

KARIUS

BASECARE

Personalis

VIOME

Synthetic biology & CRISPR

GINKGO

zymergen

TWIST

SYNTHEGO

geno

viridos

DNASCRIPT

Bluepha

TelesisBio

horizon

AI-driven drug discovery

AbCellera

BrCO

BenchSci

BigHot

Alloy Therapeutics

CytoReason

BioMap

Helixon

C4XD

koneksa

Advanced biopharma services

Sterling

SeluxDX

Ncardia

Prolific Machines

MEDx

LYVGEN

ADVANCED MEDICINE PARTNERS

Comera

LETO

XCELL

3 Emerging

Multimomics & proteomics

GRAIL

somalogic

PhenomeX

isoplexis

Thrive

NAUTILUS

emulate

epic sciences

cellanome

Detect

Microfluidics & biosensors

NUPROBE

Deepcell

glympse bio

THINKCYTE

EVEREST DETECTION

MERCY

insphero

Curate

MIROCULUS

康杰生物

Generative AI & quantum computing

XtalPi

Profluent

Facible

Quantum Biosystems

Qubit

PromiseBio

永木分子

DeepLife

QSIMULATE

Cell & gene therapy services

elevatebio

Humacyte

Rapidmicro

Probio Technology

amplifybio

TOUCHLIGHT

MaxCyte

VIROCELL BIOLOGICS

elektrofi

brigene

Q1 2025 Pharma Biotools VC Trends

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VC activity

Quarter highlights

Funding activity sees a strong rebound: Investment in pharma biotools has mirrored that of broader biopharma cycles. After peaking in 2020-2021 at \$10.4 billion across 430 deals, funding corrected through 2022-2023 before beginning to resurge in 2024. Q1 2025 demonstrated a strong rebound, recording 65 deals (up 8.3% from Q4 2024) with \$1.3 billion in funding. This marked an 81.4% quarterly increase and the highest level of investment since Q2 2024. However, the industry is very consolidated, with fewer but massive deals driven by intellectual property and patents.

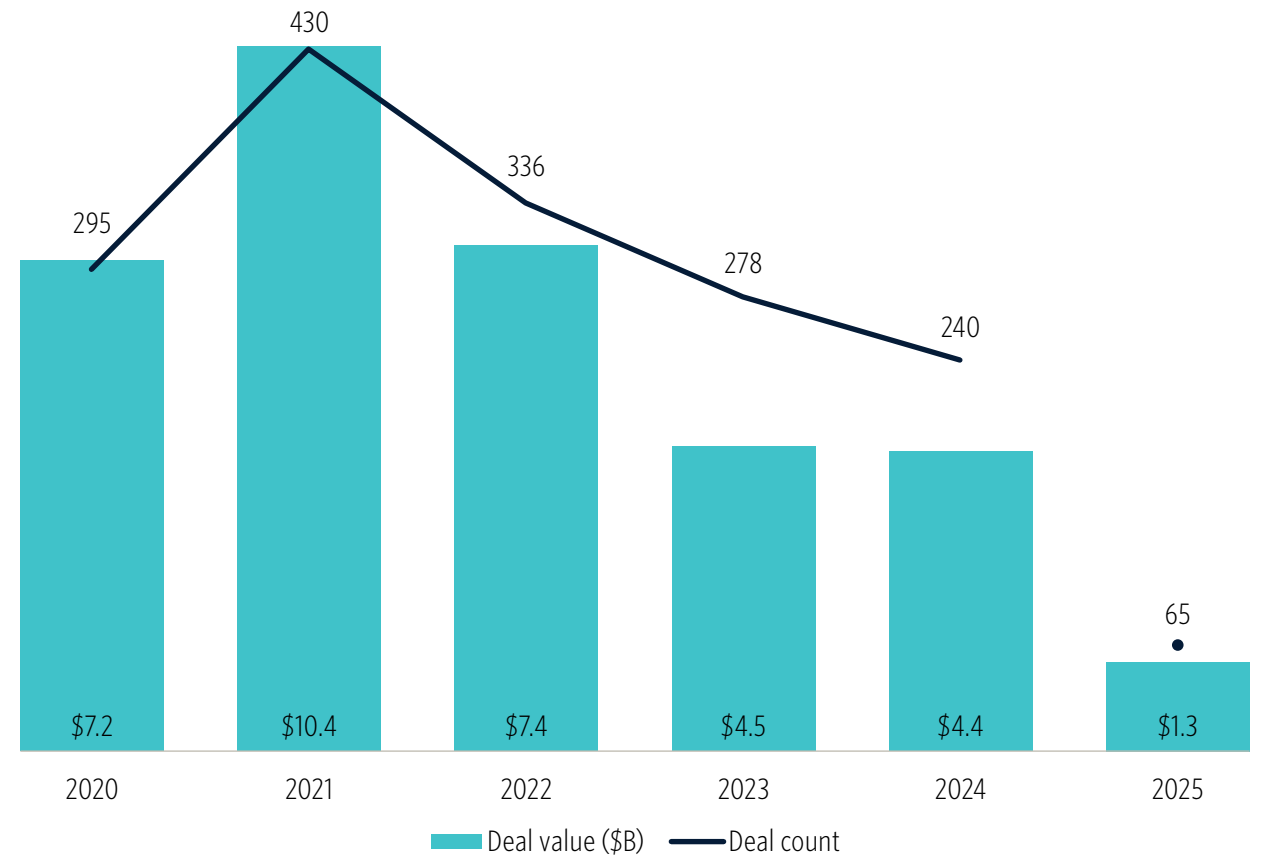
Investors shift toward higher-value deals: In the trailing 12 months, deal activity declined 9.5% to 238 deals, but total funding increased 12.8% to \$4.8 billion, indicating larger average deal sizes and investor focus on more mature opportunities. This trend suggests market stabilization around selective, quality-focused investments while maintaining robust capital deployment.

Investment trends and market dynamics

AI and computational biology lead investment focus: AI-driven drug discovery, synthetic biology, and next-generation sequencing remain top targets, with computational biology platforms seeing particularly strong interest. Investors increasingly favor multiapplication platforms over single-product plays, driving consolidation around companies with broad technology applicability.

Emerging technology areas gain momentum: Lab automation and computational tools have continued to attract sustained interest, while diagnostic multiomics platforms have gained significant momentum as personalized medicine adoption accelerates. Spatial biology, single-cell analysis tools, and AI-powered laboratory management systems are attracting growing investor attention as the industry pushes toward more integrated, data-driven research environments.

Pharma biotools VC deal activity



Source: PitchBook • Geography: Global • As of March 31, 2025



VC ACTIVITY

Strategic consolidation continues: Mature platforms continue to draw strategic M&A from industry giants such as Thermo Fisher Scientific and Danaher Corporation (exemplified by Thermo Fisher’s acquisition of Olink and Danaher’s acquisition of Abcam), while advanced and emerging technologies secure the majority of new venture dollars.

Notable Q1 2025 VC deals

Megarounds of \$100 million or more dominate funding:

- Colossal Biosciences raised a \$200 million Series C for its CRISPR-based de-extinction and synthetic biology platform, reaching a \$10.2 billion post-money valuation.
- Harrison.ai closed a \$111.8 million Series C for its AI-enabled healthcare-platform integration tools.
- Geneoscopy completed a \$105 million Series C for its digestive disease diagnostics technology, achieving a \$440 million post-money valuation.

AI-driven platforms attract significant capital:

- Chai Discovery secured a \$60 million Series A1 for its molecular-structure prediction platform, reaching a \$570 million post-money valuation.
- Bioptimus attracted a \$41.4 million Series A to develop its universal AI foundation model for biology.

- Latent Labs garnered a \$39.9 million Series A for continuing work on its AI-driven generative models for programmable biology.
- Zafrens obtained a \$23 million Series A for its cell-drug interaction mapping technology.

Specialized technology segments show strong activity:

- Stellaromics completed a \$80 million Series B to research spatial biology and gene expression analysis technology.
- Quibim closed a \$50 million Series A to design multiomics analytical tools for precision health imaging.
- Atrandi Biosciences raised a \$25 million Series A for its microfluidics technology platform.
- Wheeler Bio secured a \$36 million Series A1 for its biomanufacturing services.
- Exothera garnered \$33.2 million for its research in viral vector manufacturing services.

Exit activity and M&A

Exit activity remains at low levels: Q1 2025 recorded six exits generating \$0.2 billion in total value, consistent with the subdued exit environment that has persisted since 2022. The quarter included one public listing and five acquisitions, a continuation of the correction after 2021, when exits peaked at \$47.3 billion across 46 transactions.



VC ACTIVITY

Strategic acquisitions dominate the exit landscape: Notable exits included chromatography product developer Sepax Technologies' \$217 million IPO (the quarter's largest exit) and the strategic acquisitions of five companies: SeQure DX (acquired for \$7 million by MaxCyte), Eagle Genomics, Asia Genomics, Coriell Life Sciences, and Molecular Assemblies (acquired by Maravai LifeSciences). Most transactions involved undisclosed terms, suggesting smaller deal sizes and strategic rather than financial buyer interest.

Consolidations focus on complementary technologies: Strategic buyers continue focusing on genomics, diagnostics, and analytical tools that enhance existing product portfolios. The low exit volume indicates continued pressure on liquidity events, with strategic M&A providing the primary path to exit for biotools companies. Exit activity remains well below the sector's historical peak between 2020 and 2022, when quarterly exit values regularly exceeded \$8 billion to \$47 billion.

2025 outlook

Capital allocation trends favor proven platforms: Biotools capital allocation is expected to continue consolidating into fewer but larger deals, with investors prioritizing platform technologies with validated performance data and clear market adoption pathways. This reflects the sector's maturation and increasing focus on companies with proven commercial traction, particularly favoring AI-driven platforms and multiapplication tools over single-product solutions.

Mixed market sentiment emerges amid macroeconomic headwinds: The [2025 J.P. Morgan Healthcare Conference](#) initially set an optimistic tone, driven by renewed interest in life sciences tool innovation and AI-powered drug discovery advances. However, persistent macroeconomic pressures and funding constraints in academic markets continue directing investor focus toward established companies with diversified customer bases and recurring revenue models.

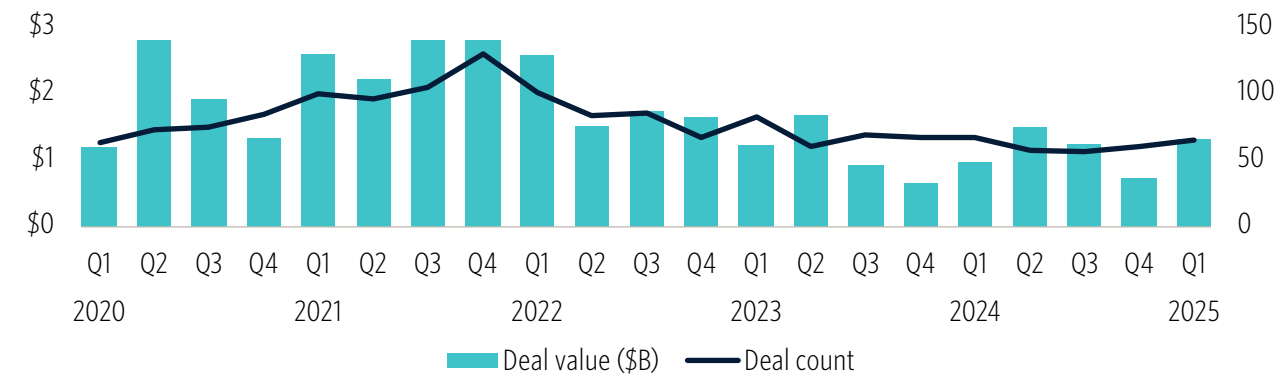
Strategic M&A: Opportunities for acquisitions are expected to grow throughout 2025, as large life sciences tool companies such as Thermo Fisher Scientific and Danaher Corporation may seek to expand their portfolios with innovative technologies. The prevailing funding environment could present attractive acquisition prospects at more reasonable valuations. Additionally, strategic buyers could utilize ample cash reserves to bolster technological capabilities during this period.

Early-stage funding remains constrained: Seed and early-stage biotools investments face significant headwinds as venture funds maintain capital discipline, primarily directing allocation to follow-on investments in proven platform companies rather than new opportunities. The challenging funding environment at academic institutions, combined with weak exit activity and limited IPO opportunities, will likely constrain capital recycling in the early-stage segment throughout 2025.



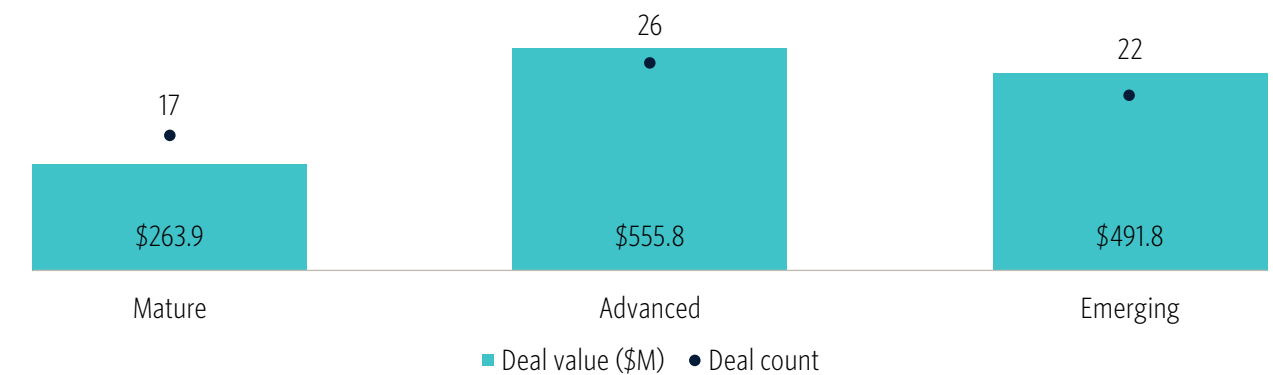
VC ACTIVITY

Pharma biotools VC deal activity by quarter



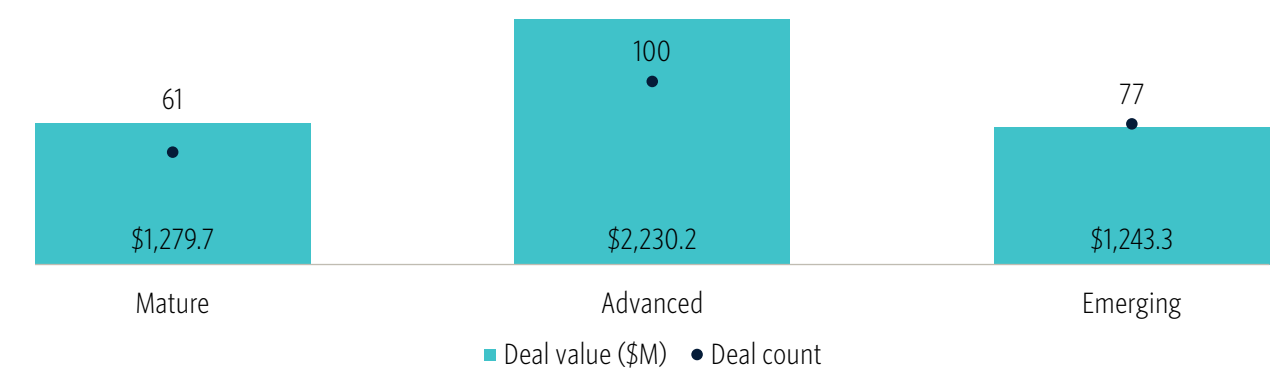
Source: PitchBook • Geography: Global • As of March 31, 2025

Q1 2025 pharma biotools VC deal activity by segment



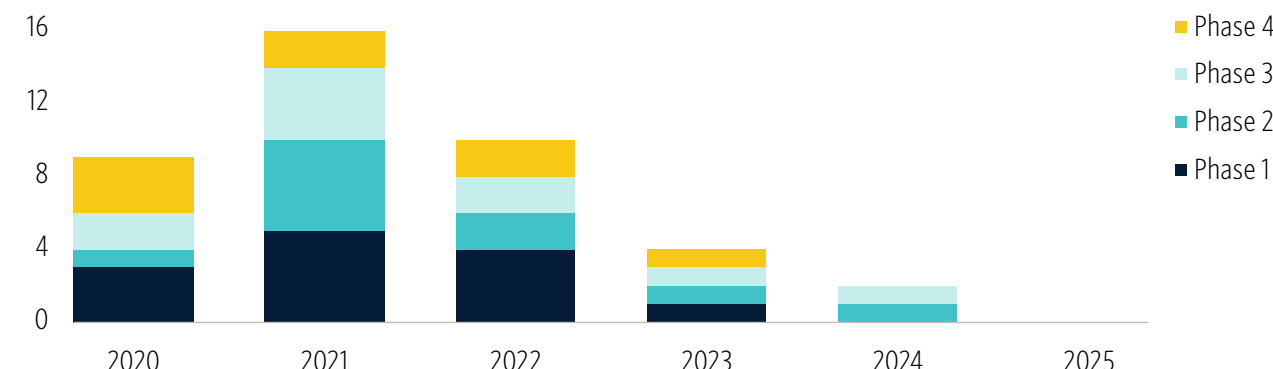
Source: PitchBook • Geography: Global • As of March 31, 2025

Trailing 12-month (TTM) pharma biotools VC deal activity by segment



Source: PitchBook • Geography: Global • As of March 31, 2025

Pharma biotools VC deal count by highest phase

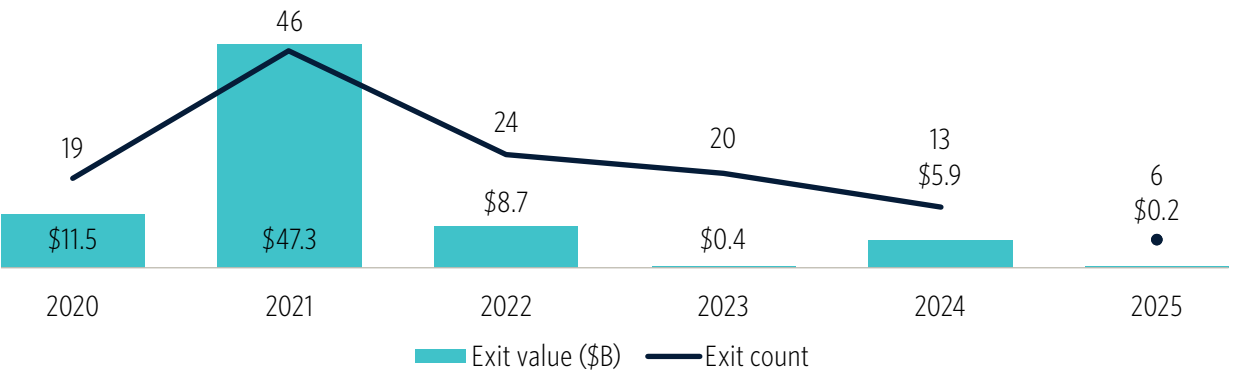


Source: PitchBook • Geography: Global • As of March 31, 2025



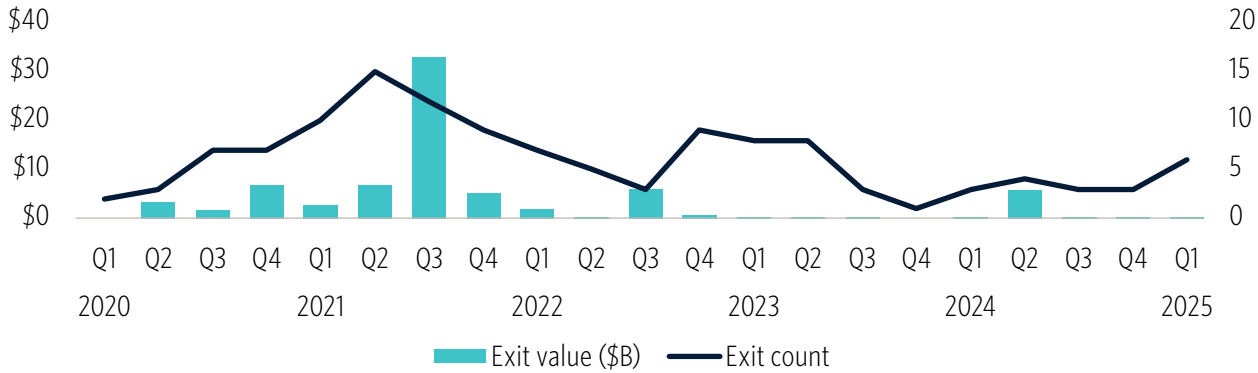
VC ACTIVITY

Pharma biotools VC exit activity



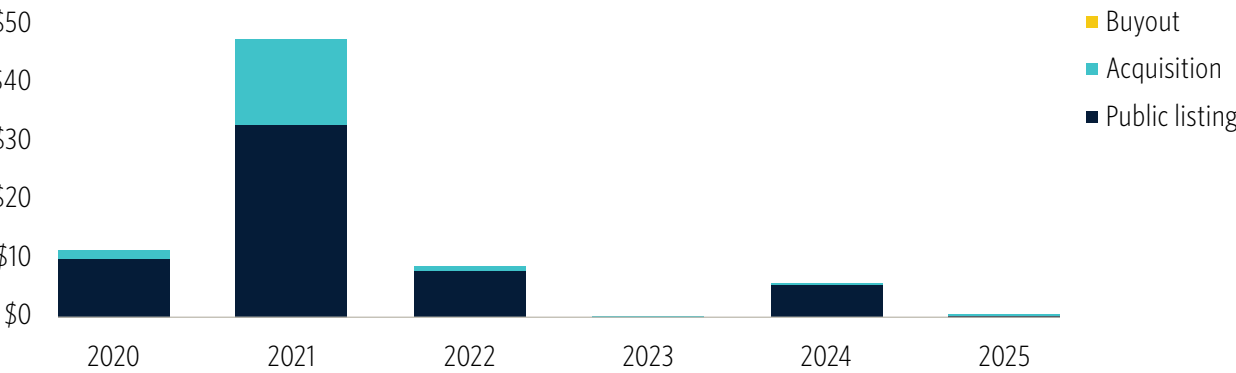
Source: PitchBook • Geography: Global • As of March 31, 2025

Pharma biotools VC exit activity by quarter



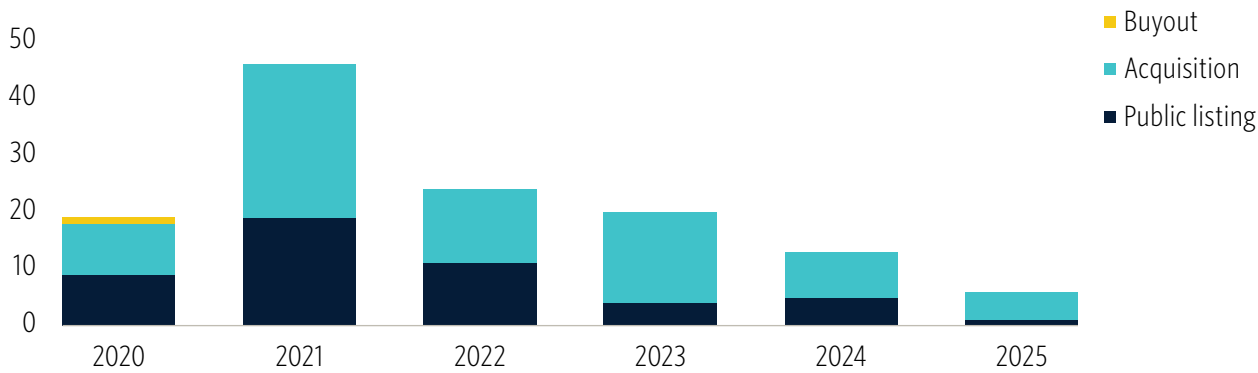
Source: PitchBook • Geography: Global • As of March 31, 2025

Pharma biotools VC exit value (\$M) by type



Source: PitchBook • Geography: Global • As of March 31, 2025

Pharma biotools VC exit count by type



Source: PitchBook • Geography: Global • As of March 31, 2025



VC ACTIVITY

Median pharma biotools pre-seed/seed deal value and pre-money valuation (\$M) by segment

	2020		2021		2022		2023		2024		2025	
Segment	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation
Mature	\$5.0	N/A	\$5.6	\$15.0	\$2.6	\$11.6	N/A	N/A	N/A	N/A	N/A	N/A
Advanced	\$2.5	\$8.0	\$3.1	\$9.5	\$5.0	\$14.4	\$5.3	\$11.3	\$7.1	\$15.0	N/A	N/A
Emerging	\$2.0	\$4.3	\$2.0	\$6.5	\$5.0	\$9.0	\$7.2	N/A	\$5.9	N/A	N/A	N/A

Source: PitchBook • Geography: Global • As of March 31, 2025



VC ACTIVITY

Median pharma biotools early-stage VC deal value and pre-money valuation (\$M) by segment

	2020		2021		2022		2023		2024		2025	
Segment	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation
Mature	\$12.1	\$34.0	\$15.8	\$40.0	\$14.8	\$24.2	\$14.5	N/A	\$25.5	N/A	N/A	N/A
Advanced	\$9.5	\$25.5	\$12.8	\$32.5	\$14.9	\$27.5	\$10.2	\$49.4	\$13.0	\$52.6	\$20.7	N/A
Emerging	\$8.7	\$28.7	\$15.0	\$27.0	\$11.0	\$28.9	\$22.7	\$30.0	\$14.9	N/A	N/A	N/A

Source: PitchBook • Geography: Global • As of March 31, 2025



VC ACTIVITY

Median pharma biotools late-stage VC deal value and pre-money valuation (\$M) by segment

	2020		2021		2022		2023		2024		2025	
Segment	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation
Mature	\$14.1	\$54.2	\$15.7	\$164.0	\$15.0	\$64.0	\$10.0	\$48.9	\$7.0	\$35.0	\$13.8	N/A
Advanced	\$17.6	\$55.0	\$14.0	\$48.9	\$23.0	\$47.5	\$11.7	\$52.2	\$15.0	\$76.7	\$4.9	N/A
Emerging	\$9.2	\$46.2	\$16.2	\$47.9	\$8.8	\$57.6	\$7.8	\$34.4	\$6.0	\$15.7	\$14.7	N/A

Source: PitchBook • Geography: Global • As of March 31, 2025



VC ACTIVITY

Median pharma biotools venture growth deal value and pre-money valuation (\$M) by segment

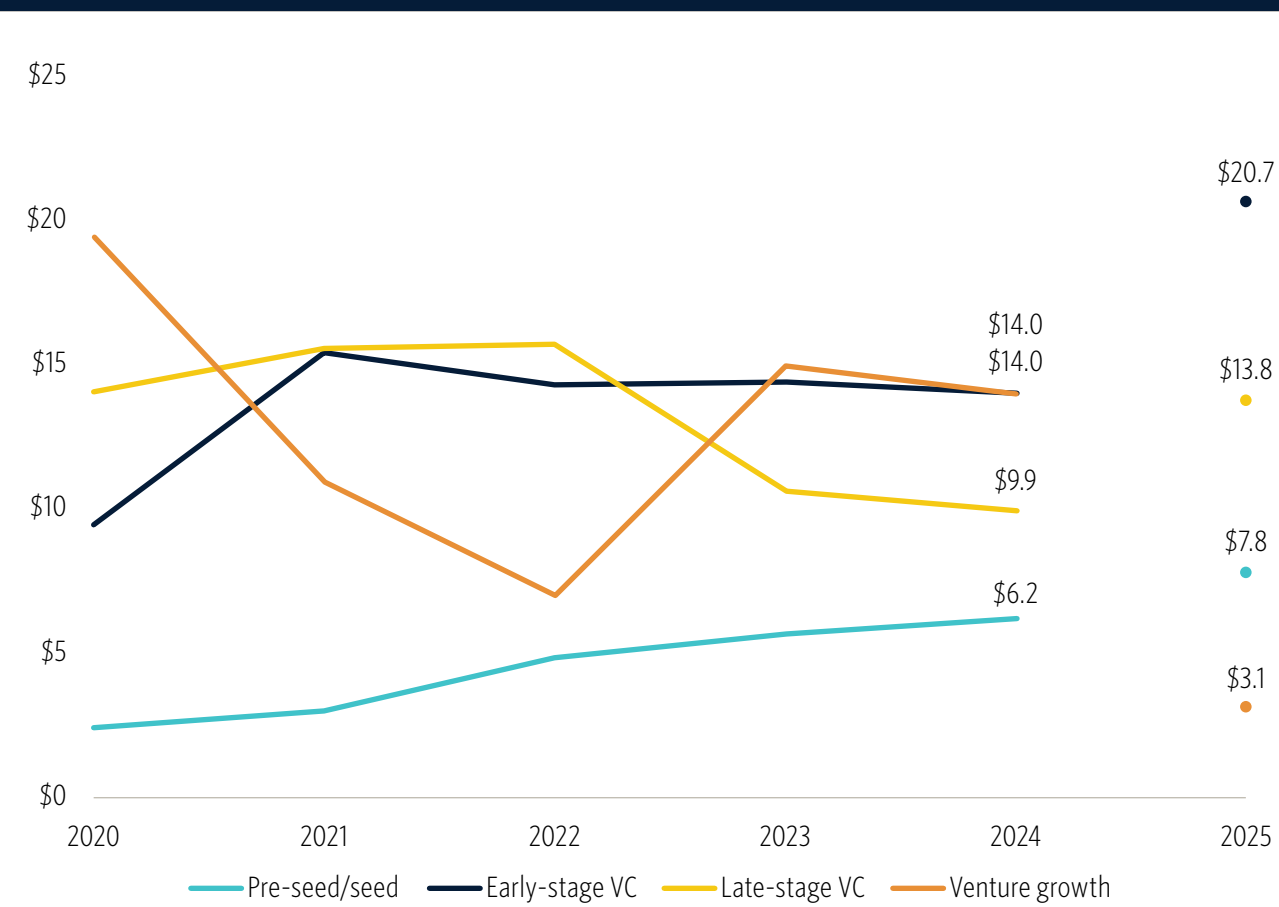
	2020		2021		2022		2023		2024		2025	
Segment	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation	Deal value	Pre-money valuation
Mature	\$9.2	\$44.9	\$8.0	\$26.0	N/A	N/A	\$14.3	N/A	\$11.4	N/A	N/A	N/A
Advanced	\$70.0	N/A	\$95.4	\$187.3	\$20.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Emerging	N/A	N/A	\$9.0	N/A	N/A	N/A	N/A	N/A	\$27.5	N/A	N/A	N/A

Source: PitchBook • Geography: Global • As of March 31, 2025



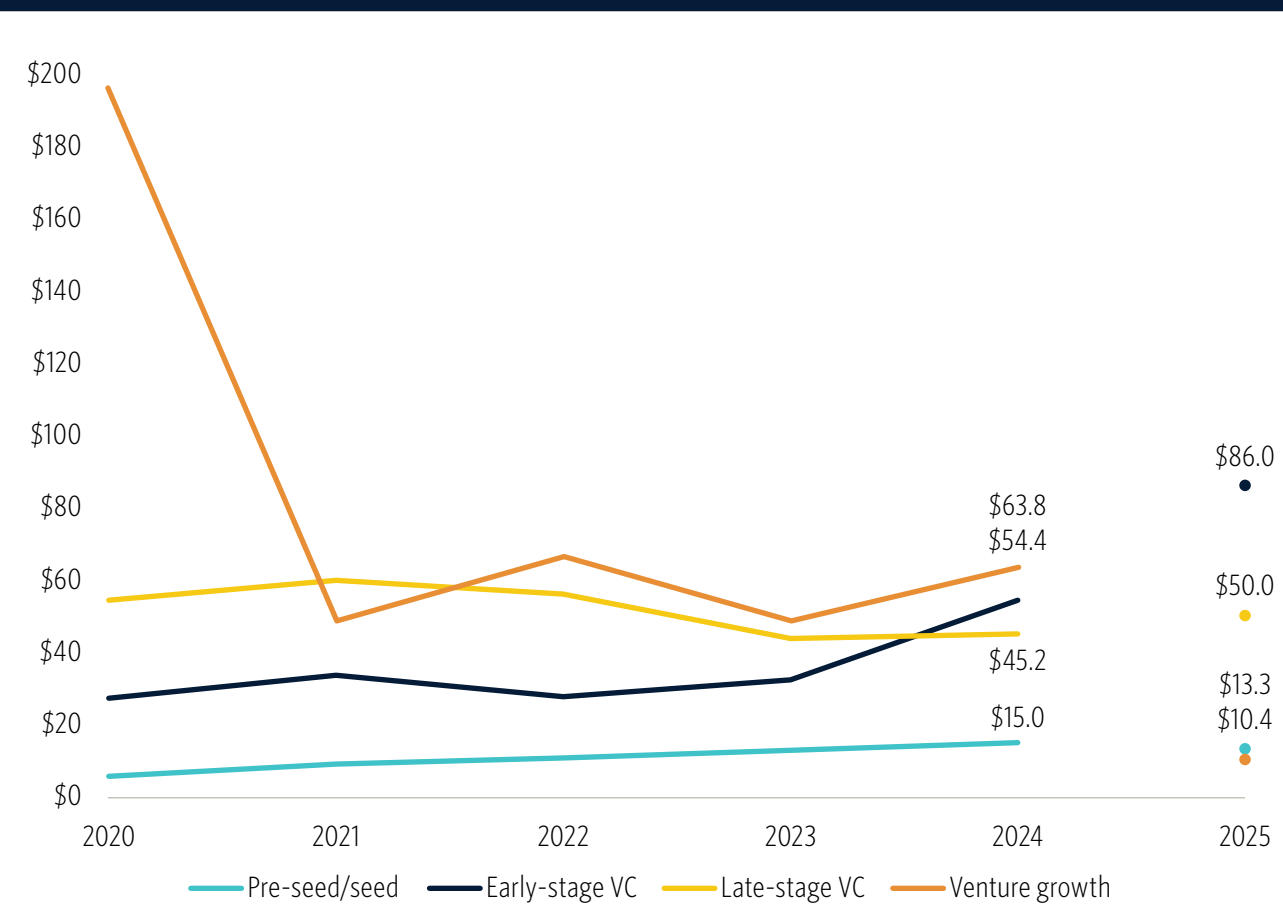
VC ACTIVITY

Median pharma biotools VC deal value (\$M) by stage



Source: PitchBook • Geography: Global • As of March 31, 2025

Median pharma biotools VC pre-money valuation (\$M) by stage



Source: PitchBook • Geography: Global • As of March 31, 2025



VC ACTIVITY

Pharma biotools heatmap by segment

Segment	TTM total deal value (\$M)	TTM total deal count	YoY deal value growth	YoY deal count growth
Advanced	\$2,230.2	100	102.7%	6.4%
Emerging	\$1,243.3	77	-35.4%	-14.4%
Mature	\$1,279.7	61	5.2%	-23.8%

Source: PitchBook • Geography: Global • As of March 31, 2025



VC ACTIVITY

Key pharma biotools early-stage VC deals in Q1 2025

Company	Close date	Deal value (\$M)	Category	Lead investor(s)	Valuation step-up (post to pre)
Inhal Pharma	March 12	N/A	Traditional manufacturing, bioprocessing & pharma services	N/A	N/A
Infinity Bio	March 11	\$2.0	Basic molecular & cell biology	N/A	N/A
Synvect	March 11	\$3.0	Synthetic biology & CRISPR	N/A	N/A
InnoGene BioTech	March 3	\$20.7	Synthetic biology & CRISPR	Hongsheng Capital, Zhangjiang Hi-Tech Investment	N/A
Ensocell	February 24	\$13.1	Multimomics & proteomics	N/A	1.0x
Persist AI	February 14	\$12.0	Basic molecular & cell biology	N/A	N/A
Stellaromics	February 11	\$80.0	Multimomics & proteomics	Bayshore Global Management, Catalyst4	1.3x
Zafrens	February 5	\$23.0	AI-driven drug discovery	Prime Movers Lab	1.4x
Nucleus	January 30	\$14.0	Genomics & sequencing	N/A	N/A
Matwings Technology	January 23	N/A	AI-driven drug discovery	N/A	N/A

Source: PitchBook • Geography: Global • As of March 31, 2025



VC ACTIVITY

Key pharma biotools late-stage VC deals in Q1 2025

Company	Close date	Deal value (\$M)	Category	Lead investor(s)	Valuation step-up (post to pre)
Octave	March 22	\$31.8	AI-driven drug discovery	N/A	0.3x
Wheeler Bio	March 20	\$36.0	Traditional manufacturing, bioprocessing & pharma services	Alloy Therapeutics, Echo Investment Capital	N/A
Huixin Lifetech	March 5	\$13.8	Microfluidics & biosensors	N/A	N/A
Kryptos Biotechnologies	February 20	\$12.0	Basic molecular & cell biology	OSANG Healthcare	1.0x
Atrandi Biosciences	February 18	\$25.0	Microfluidics & biosensors	Lux Capital	N/A
Phlow	February 14	\$30.5	Traditional manufacturing, bioprocessing & pharma services	N/A	1.9x
Harrison.ai	February 12	\$111.8	AI-driven drug discovery	Aware Super, Horizons Ventures	N/A
Agile Bio	February 12	\$13.8	Traditional manufacturing, bioprocessing & pharma services	N/A	N/A
Transcend Vivoscope	February 12	\$13.8	Analytical & pharma tools	Fosun Health Capital, Shenzhen Capital Group	N/A
Biolog	February 10	\$16.1	Basic molecular & cell biology	N/A	N/A

Source: PitchBook • Geography: Global • As of March 31, 2025



VC ACTIVITY

Key pharma biotools VC exits in Q1 2025

Company	Close date	Exit value (\$M)	Category	Acquirers(s)	Exit type
Sepax Technologies	January 10	\$216.9	Analytical & pharma tools	Citic Securities Investment	IPO
SeQure DX	January 29	\$7.0	Genomics & sequencing	MaxCyte	M&A
Eagle Genomics	January 10	N/A	Bioinformatics & cheminformatics	N/A	M&A
Asia Genomics	January 1	N/A	Genomics & sequencing	Glow	M&A
Coriell Life Sciences	March 19	N/A	Genomics & sequencing	InformedDNA	M&A
Molecular Assemblies	January 28	N/A	Synthetic biology & CRISPR	Maravai LifeSciences	M&A

Source: PitchBook • Geography: Global • As of March 31, 2025



VC ACTIVITY

Top strategic acquirers of pharma biotools companies since 2020

Acquirer	Deal count	Investor type
Illumina	3	Corporation
Recursion Pharmaceuticals	3	Corporation
Exact Sciences	3	Corporation
Ginkgo Bioworks	2	Corporation
Bruker	2	Corporation
Bio-Rad Laboratories	2	Corporation
BICO Group	2	Corporation
Charles River Laboratories International	2	Corporation

Source: PitchBook • Geography: Global • As of March 31, 2025

Top VC investors in pharma biotools companies since 2020

Company	Total deal count	Pre-seed/seed	Early-stage VC	Late-stage VC	Venture growth	Investor type
Northpond Ventures	29	3	6	15	5	VC
Legend Capital	21	0	11	9	1	VC
Casdin Capital	21	1	8	11	1	VC
GL Ventures	18	0	10	8	0	VC
F-Prime Capital	18	0	5	12	1	VC
Hongshan Capital Group	18	2	10	5	1	VC
SOSV	18	7	4	7	0	VC
Khosla Ventures	16	5	3	8	0	VC
Alumni Ventures	16	4	5	7	0	VC

Source: PitchBook • Geography: Global • As of March 31, 2025



VC ACTIVITY

Top VC-backed pharma biotools companies by total VC raised to date

Company	VC (\$M) raised to date	Segment	Category	IPO probability	M&A probability	No exit probability
ElevateBio	\$1,269.0	Emerging	Cell & gene therapy services	67%	31%	2%
Formation Bio	\$618.8	Mature	Traditional manufacturing, bioprocessing & pharma services	92%	5%	3%
Human Longevity	\$587.8	Advanced	Genomics & sequencing	89%	9%	2%
DNAexus	\$568.7	Advanced	Genomics & sequencing	84%	14%	2%
EirGenix	\$479.6	Mature	Traditional manufacturing, bioprocessing & pharma services	N/A	N/A	N/A
Synthego	\$459.5	Advanced	Synthetic biology & CRISPR	97%	1%	2%
Colossal Laboratories & Biosciences	\$437.6	Advanced	Synthetic biology & CRISPR	90%	5%	5%
Delfi Diagnostics	\$431.6	Advanced	Genomics & sequencing	62%	36%	2%
Probio Technology	\$411.3	Emerging	Cell & gene therapy services	23%	71%	6%
Geno	\$401.5	Advanced	Synthetic biology & CRISPR	37%	61%	2%

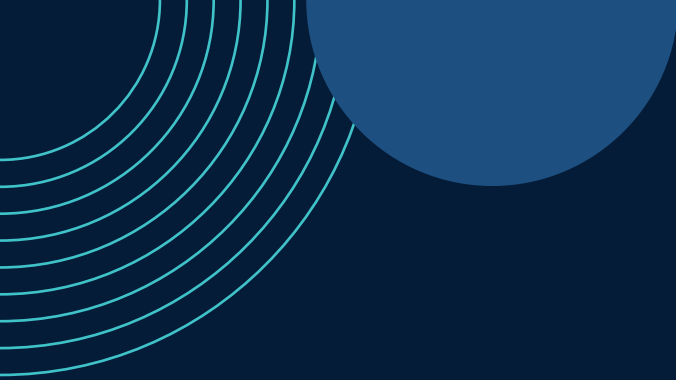
Source: PitchBook • Geography: Global • As of March 31, 2025
Note: Probability data is based on [PitchBook VC Exit Predictor methodology](#).



Pharma biotools VC deal summary

	Quarterly activity					Trailing 12-month activity	
	Q1 2024	Q2 2024	Q3 2024	Q4 2024	Q1 2025	Q2 2023 to Q1 2024	Q2 2024 to Q1 2025
Deal count	67	57	56	60	65	263	238
QoQ change	0.0%	-14.9%	-1.8%	7.1%	8.3%	N/A	-9.5%
Share of total VC	0.6%	0.6%	0.6%	0.7%	0.8%	0.6%	0.6%
Deal value (\$B)	\$1.0	\$1.5	\$1.2	\$0.7	\$1.3	\$4.2	\$4.8
QoQ change	44.9%	55.4%	-17.2%	-41.3%	81.4%	N/A	12.8%
Share of total VC	1.1%	1.6%	1.5%	0.6%	0.9%	1.3%	1.1%
Exit count	3	4	3	3	6	15	16
Public listings	0	3	1	1	1	4	6
Acquisitions	3	1	2	2	5	11	10
Buyouts	0	0	0	0	0	0	0

Source: PitchBook • Geography: Global • As of March 31, 2025



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Our Industry and Technology Research provides detailed analysis of established industries and nascent tech sectors from the perspective of private market dealmaking, helping you stay current on market trends and providing the insights you need to pursue new opportunities with confidence.

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